Reproducing "Towards Safer Pretraining": Public Artifacts Replicate, Closed APIs Diverge

Anonymous authors
Paper under double-blind review

Abstract

We reproduce four core claims from Mendu et al. (2025)'s framework for harmful content detection in web-scale pretraining data, using only publicly released artifacts. The HAVOC benchmark replicates near-exactly (26.76% leakage versus 26.7% reported, $\Delta = 0.06$ pp). The TTP prompt-based classifier, by contrast, shows a 21-point F1 gap on TTP-Eval when first measured (0.62 vs. 0.83) and a 25-point gap on OpenAI Moderation (0.55 vs. 0.80), both driven by lower recall at comparable precision. Re-runs on byte-identical code two weeks later close both gaps ($F_1 \in [0.78, 0.79]$ on TTP-Eval across two endpoints; $F_1 = 0.83$ on OpenAI Moderation), isolating silent `gpt-4o` snapshot drift as the dominant cross-benchmark source of variance and itself constituting a reproducibility finding. HarmFormer evaluated on TTP-Eval reaches 0.78 F1, a number the original paper does not report and that we contribute as a public-artifact baseline. We complement the reproduction with two additional experiments along axes the original authors leave open. (i) Cross-model TTP generalization on five additional LLMs (Gemma 2/3 27B, GPT-OSS 20B, R1-Distill-Qwen-32B, Gemini 2.0 Flash) shows the prompt is portable: DeepSeek-R1 matches GPT-4o (0.63 vs. 0.62 F1) and Gemini outperforms it (0.80 F1). A full $n = 393$ raw-output audit on the four open-weight rows reveals that parser failures are negligible (0–6 of 393 per model) and that cross-model F1 variance (0.35–0.60) maps almost monotonically onto each model's rate of emitting the toxic (Intent) label (1.4–5.4%) — a labelling-policy effect, not a parser-format effect. (ii) Multilingual evaluation across six languages on NLLB-200 translations of TTP-Eval shows HarmFormer's F1 drops from 0.78 (English) to 0.21–0.40, with no consistent script or family pattern, while Llama Guard's translated-language F1s straddle its own English baseline (0.36) and so are dominated by a benchmark-mismatch effect rather than a language effect. Taken together, taxonomy-driven harm detection is reproducible where artifacts are released and inference is local (HAVOC, HarmFormer), but fragile where it depends on closed APIs (TTP) or English-only training data (HarmFormer multilingual). All experiments are tracked in Weights & Biases with CodeCarbon energy traces, and the code and run links are released for independent verification.

1 Introduction

Large language models (LLMs) are increasingly pretrained on massive web-scale datasets, raising growing concerns about the presence of harmful, toxic, or manipulative content in their training data (Bender et al., 2021). Prior work has shown that large web corpora often lack sufficient documentation and transparency, making it difficult to assess data quality and potential risks introduced during pretraining (Dodge et al., 2021). Moreover, evaluating and mitigating harmful content frequently relies on automatic or model-based judgments, which are known to be imperfect and sensitive to evaluation design choices (Welbl et al., 2021). Since pretraining data directly shapes downstream model behavior, understanding and mitigating harmful content has become a central issue in responsible AI development.

The IJCAI 2025 paper *Towards Safer Pretraining: Analyzing and Filtering Harmful Content in Webscale Datasets for Responsible LLMs* addresses this challenge by proposing a taxonomy-driven framework for analyzing harmful content in web data and by introducing tools to support safer pretraining (Mendu et al., 2025). The authors define a structured taxonomy that distinguishes between safe, topical, and toxic content, building on earlier efforts to study toxic language in language model outputs and training data (Gehman et al., 2020). In addition, the paper introduces several resources, including the TTP benchmark for prompt-based harmful content identification, the HAVOC benchmark for adversarial prompt evaluation, and the HarmFormer model for long-form harmful content classification. These contributions align with a broader line of work on adversarial prompting and robustness benchmarking.

In this work, we examine the reproducibility of the results reported in the original paper, focusing on the four core claims for which released artifacts make direct verification possible: TTP performance on TTP-Eval, HarmFormer performance, baseline comparisons on the OpenAI Moderation set, and HAVOC leakage. While the authors release prompt templates, model weights, benchmark datasets, and pre-computed model generations, the large-scale data processing pipeline (3M-page TTP labeling, internal HarmFormer test split, 1M-sample prevalence study) is described at a high level only and is not reproducible without comparable budget and access. Rather than treat this as a barrier, we report what does and does not replicate from the public surface and characterize which discrepancies stem from artifact ambiguity versus from the underlying methodology.

We organise this verification around the four pillars of responsible AI that scaffold this study—fairness, accountability, confidentiality, and transparency (FACT)—and treat each pillar as a concrete check on the released framework rather than as a generic disclaimer. *Accountability* requires that headline performance numbers be independently verifiable; we therefore re-evaluate every reproducible claim from a clean checkout of the released artifacts and report a per-claim status table (Table 4). *Transparency* requires that the path from raw data to reported number be auditable; we log every experiment to Weights & Biases with CodeCarbon energy traces and ship the Slurm scripts that produced each row of every table. *Fairness* requires that the safety surface a classifier exposes be roughly even across the populations it is applied to; the multilingual experiment in Section 7 measures whether the released English classifiers extend to non-English subsets of the same web corpora they are deployed against. *Confidentiality* is partially in scope: we operate exclusively on already-public artifacts and do not redistribute private user data, but we discuss in Section 9 where the released benchmarks themselves contain content whose handling deserves more care than the original paper specifies. These four pillars motivate our reading of the results and structure the discussion in Section 8.

Our specific contributions are: **(i)** a per-claim reproducibility table (Table 4) showing that two of four claims reproduce within tolerance from a clean checkout, two TTP claims diverge by more than 20 F1 points in our original April measurement and then close on May re-runs (TTP-Eval: $F_1 \in [0.78, 0.79]$; OpenAI Moderation: $F_1 = 0.83$), surfacing silent `gpt-4o` snapshot drift as a cross-benchmark phenomenon, and one is partially validated against a publicly available proxy benchmark; **(ii)** a public-artifact HarmFormer baseline on TTP-Eval (0.78 F1), which the original paper does not publish and which is needed for any future cross-model comparison; **(iii)** a cross-model TTP generalization study on five additional LLMs that disentangles prompt-format failures from semantic ones; **(iv)** a multilingual robustness study covering six languages and four scripts, addressing the multilingual gap that the original paper explicitly defers to future work; and **(v)** fully tracked experiments (Weights & Biases run links, CodeCarbon energy traces, Slurm job scripts) released alongside the code so that every reported number can be regenerated by an independent reader.

2 Related Work

Prior work related to this study spans research on web-scale dataset transparency, harmful content detection tools, and toxicity evaluation benchmarks.

2.1 Web-Scale Data and Transparency

Web-scale datasets are widely used for pretraining language models, but their opacity raises concerns about data quality, bias, and downstream harms. Dodge et al. (2021) document limitations in dataset transparency for corpora such as C4, motivating clearer reporting of data curation practices. Luccioni & Viviano (2021) analyze undesirable content in Common Crawl using keyword heuristics, while Jansen et al. (2022) propose perplexity-based filtering to remove toxic text. However, these methods lack precision: keyword filters can conflate harmful intent with educational content, and perplexity thresholds fail to capture contextual nuances.

2.2 Content Moderation Tools

Existing approaches to harmful content detection rely on keyword blocklists (such as the LDNOOBW list[1]) and API-based tools. Perspective API[2] is widely used for toxicity detection but suffers from critical limitations: it is designed for sentence-level analysis and uses binary taxonomies that conflate harmful intent with socially critical discourse (Mendu et al., 2025). Model-based approaches include HateBERT (Caselli et al., 2021), a BERT-based model fine-tuned on hate speech datasets, and Llama Guard (Inan et al., 2023), an LLM-based input-output safeguard for conversational AI. The original authors demonstrate that these tools exhibit poor generalization across harm categories and struggle with long-form content, motivating their three-dimensional taxonomy (Safe, Topical, Toxic) and multi-harm coverage.

2.3 Toxicity Evaluation Benchmarks

RealToxicityPrompts (RTP) (Gehman et al., 2020) evaluates toxic degeneration in language model outputs using 100K prompts scored by Perspective API. However, RTP focuses narrowly on toxicity and sexual content, and subsequent work shows that automatic toxicity assessment is sensitive to model choice and evaluation design (Welbl et al., 2021). To address these limitations, the original authors introduce HAVOC (Harmful Abstractions and Violations in Open Completions), a multidimensional benchmark with 10,376 adversarial prompts spanning five harm categories. Unlike RTP, HAVOC distinguishes between neutral, passive (topical), and provocative inputs, revealing that state-of-the-art LLMs exhibit harmful content in 26.7% of outputs.

2.4 Contributions of the Original Work

The work of Mendu et al. (2025) integrates structured taxonomies with prompt-based (TTP) and model-based (HarmFormer) tools for large-scale harmful content analysis. Our study focuses on assessing the reproducibility of these contributions using the publicly released artifacts, including the TTP prompt templates, HAVOC benchmark, TTP-Eval evaluation set, and the HarmFormer model weights.

3 Scope of Reproducibility

We verify four claims from Mendu et al. (2025) that are testable from the released artifacts: **Claim 1 (TTP Performance)**: 0.87 precision, 0.79 recall, 0.83 F1 on TTP-Eval (Original Table 3); **Claim 2 (HarmFormer Performance)**: 0.88 precision, 0.81 recall, 0.85 F1 on the authors' internal split (Original Table 6), noting we instead evaluate on TTP-Eval; **Claim 3 (Baseline Comparison)**: TTP and HarmFormer outperform baselines on OpenAI Moderation (Original Table 7), evaluated here against Perspective and Llama Guard with TTP via OpenRouter (gpt-4o); and **Claim 4 (HAVOC Leakage)**: 26.7% overall leakage with 76% on provocative inputs (Original Table 10, Original Figure 3), computed from released `havoc_modeleval.tsv` generations.

All results in this paper are generated from our reproduction code (anonymized: `https://anonymous.4open.science/r/TowardsSaferPretraining-6046`).

Communication with original authors. We contacted the original authors during the project to request clarifications and missing artifacts needed for strict comparability (e.g., the internal HarmFormer test split and evaluation details). We did not receive a response by the submission deadline. Therefore, all results reported in this paper rely solely on publicly released artifacts and documentation.

[1] `https://github.com/LDNOOBW/List-of-Dirty-Naughty-Obscene-and-Otherwise-Bad-Words`
[2] `https://perspectiveapi.com/`

Due to computational and budgetary constraints, we do not reproduce the large-scale dataset annotation pipeline described in Section 5.1 of the original paper, which involves labeling 3 million web pages. We also do not reproduce the model architecture comparison experiments (Original Table 5), as we use the pre-trained HarmFormer model released by the authors rather than training from scratch.

We focus our experimental validation on reproducing Original Table 3, Original Table 4, Original Table 6, Original Table 7, and Original Table 10, and Original Figure 3 from the original paper. We do not attempt to reproduce Original Table 5 (model architecture comparison requiring retraining), Original Table 8 (3M sample dataset analysis), Original Figure 2 (domain distribution), or Original Figure 5. Original Table 1, Original Table 2, and Original Table 9 are definitional or descriptive and do not require experimental reproduction. Our code produces the reproduced tables and Original Figure 3 using the released artifacts; other figures are added to explain our additional experiments. Our reproduction therefore focuses on validating the effectiveness of the released artifacts for harmful content detection, rather than reconstructing the full data curation and model training pipeline.

4 Methodology

4.1 Model Descriptions

TTP (Topical & Toxic Prompt). TTP is a prompt-based classifier: it consists of a long instruction prompt, sent to a general-purpose LLM, that asks the LLM to score a piece of text along the original paper's three-dimensional taxonomy. The taxonomy assigns one of three labels (Safe, Topical, Toxic) to each of five harm categories—Hate & Violence, Ideological Harm, Sexual Content, Illegal Activities, and Self-Inflicted Harm—and a document is treated as Toxic if any category is Toxic. Mendu et al. tune the prompt against GPT-4o (also called GPT-4 Omni); we use the same released prompt template[3] with greedy decoding (temperature = 0) so that the only intentional difference between our runs and theirs is the API endpoint (OpenAI direct in their setup, OpenRouter in ours). For API-backed TTP runs we do not apply explicit input truncation; for the local TTP runs reported in Section 6 we cap input length at 8,192 tokens to bound KV-cache memory.

HarmFormer. HarmFormer is the open-weight, self-contained alternative to TTP. It is a fine-tuned Longformer model (Beltagy et al., 2020) with a 1024-token context window and a multi-task head: five classification heads, one per harm category, each producing a Safe / Topical / Toxic label. We use the released checkpoint[4] without further fine-tuning, and apply the same any-head-Toxic aggregation rule as TTP so the two classifiers can be compared on the same surface.

4.2 Datasets

We evaluate on three datasets from the original paper.

TTP-Eval is a 393-page TSV released by the authors, with full web-page text, harm-category labels, and Safe/Topical/Toxic annotations per category. Inter-annotator agreement is high (Krippendorff's $\alpha = 0.77$). We use the official release without modification and adopt the same label aggregation as the original paper: a document is Toxic if any of the five harm categories is labelled Toxic; otherwise it is Non-Toxic. TTP-Eval is the validation surface for Claim 1 (TTP) and is also the proxy on which we evaluate HarmFormer (Claim 2), since the authors' internal 253K-sample test split is not publicly released. The HarmFormer-on-TTP-Eval result therefore measures generalization to human-annotated data rather than reproducing Original Table 6 exactly.

HAVOC contains 10,376 prefix–suffix pairs sampled from Common Crawl, C4, and FineWeb, annotated across the five harm dimensions. Prefixes average 26 whitespace tokens (range 1–891). Approximately 17.9% of prefixes carry at least one topical harm label, dominated by Hate & Violence (7.3%); Appendix Figure A1 shows the full distribution and motivates per-category reporting. We use `havoc_modeleval.tsv`, which ships pre-computed generations and TTP judge annotations for the six models the original paper

[3] https://github.com/themendu/TowardsSaferPretraining
[4] https://huggingface.co/themendu/HarmFormer

evaluates: Gemma 2 (2B/9B/27B), Llama 3.2 (1B/3B), and Mistral 7B v0.3. We compute leakage from these annotations directly, which matches the original Original Table 10 protocol and avoids re-running inference. *Implementation note:* the released file contains a small number of rows with malformed quoting that Python's default CSV reader merges across line boundaries, silently reducing the effective sample count. We parse the file line-by-line on tab delimiters so all 10,376 rows load.

OpenAI Moderation test set (Markov et al., 2023) contains 1,680 sentences (1,158 non-toxic, 522 toxic) across Sexual, Hate, Violence, and Self-Harm. Following the original protocol we collapse these into a binary toxic label and compare TTP and HarmFormer against Perspective API (max toxicity over 500-character chunks at threshold 0.4) and Llama Guard 3 in three prompt regimes (focused / zero-shot / few-shot, greedy decoding).

4.3 Hyperparameters

We follow the original paper: TTP uses greedy decoding (temperature = 0.0, top-p = 1.0); HarmFormer uses the released checkpoint defaults (1024-token context) with no additional fine-tuning; Perspective uses max toxicity over 500-character chunks with threshold 0.4; HAVOC leakage uses the released generations (max 200 output tokens per completion).

4.4 Experimental Setup and Code

All experiments are conducted using a combination of cloud API services and academic GPU resources. For TTP evaluation, we use OpenRouter to access GPT-4o. For HarmFormer inference and for our cross-model experiment with Gemma 2 27B, we use an A100 GPU (40GB) provided by our institution's academic computing cluster. For HAVOC, unlike the original paper, which uses Ollama as the client-server for running open-source models, we do not run any model ourselves because our Slurm-based cluster environment is not compatible with Ollama, and to avoid unnecessary recomputation. Instead, we use the pre-computed generations and TTP-based annotations provided by the original paper authors to efficiently compute the leakage metrics on CPU.

For API-backed evaluations, we explicitly track parsing/format failures and apply a configurable invalid-output policy (exclude / count-as-non-toxic / count-as-toxic). Unless otherwise stated, the main tables exclude invalid outputs from metric computation; sensitivity analyses can be reproduced using the provided `-invalid-policy` flag in our scripts.

Reproducibility artifacts. Every experiment in this paper is logged to a Weights & Biases project with the same run name as the script that produced it; configuration (sanitized of API keys), per-run metrics, and the resulting JSON are uploaded as W&B artifacts so any reported number traces back to a specific commit, command, and hardware allocation. Runtime and per-run energy use are tracked with CodeCarbon 3.2.1 and exported to `results/codecarbon/`. Slurm job scripts under `jobs/` encode the exact module loads, partitions, and resource requests, and source the same shared `.env` so a reader with cluster access can rerun any experiment with a single `sbatch` command. The anonymized repository (link above) includes these scripts, the unit tests, and a quickstart that pins Python and CUDA versions.

4.5 Deviations from the Original Setup

Several pieces of the original paper's pipeline cannot be reproduced exactly from the public surface. We collect the deliberate deviations here so that each subsequent table can be interpreted against an explicit baseline.

TTP API endpoint. The paper queries GPT-4o through the OpenAI API directly. We query the same model through OpenRouter (which routes to Azure-hosted GPT-4o), since this is the API key our institution makes available. We later A/B-tested both endpoints on the full TTP-Eval (Section 5.1.1); they yielded statistically indistinguishable F_1, ruling out endpoint variance as the source of the recall gap and isolating silent `gpt-4o` snapshot drift instead.

HarmFormer test set. The original Table 6 evaluates HarmFormer on a 253K-sample internal test split that is not released. We evaluate on the released TTP-Eval (393 samples) instead and report the result as a public-artifact baseline rather than a strict reproduction.

HAVOC inference. The paper runs the six target models locally via Ollama. Our Slurm-based cluster cannot host an Ollama server, so we use the released `havoc_modeleval.tsv` (pre-computed generations and TTP judge annotations) and re-run only the leakage aggregation step. This avoids reintroducing model-version drift but means we do not independently verify the upstream generations.

Llama Guard version. The paper benchmarks against an earlier Llama Guard release. We evaluate the current public release `meta-llama/Llama-Guard-3-8B` (Llama Guard 3) because it is the gated artifact a current reader can actually request access to.

Multilingual evaluation surface. The original paper does not evaluate multilingually. Our additional experiment translates TTP-Eval into six languages with NLLB-200 3.3B and reuses the original English Safe/Topical/Toxic labels rather than re-annotating; we discuss the cost of this choice in Section 9.

Inference precision. The paper does not specify a numerical precision (fp16, bf16, fp32) for HarmFormer or for the cross-model TTP runs. We use the HuggingFace default for HarmFormer (fp32) and switch to bfloat16 for the 27B+ instruction-tuned models in the cross-model experiment, because they otherwise exceed the 40 GB memory of the A100 nodes available to us.

HAVOC CSV parser. The released `havoc_modeleval.tsv` contains a small number of rows whose embedded quotes confuse Python's default `csv` reader, silently merging rows and reducing the effective sample count. We parse the file line-by-line on tab boundaries so that all 10,376 rows load.

4.6 Computational Requirements

We estimate: TTP on TTP-Eval costs roughly \$5–8 (1̃.5M input and 100K output tokens at GPT-4o pricing); HarmFormer on TTP-Eval takes 1–2 A100 GPU hours; OpenAI Moderation baselines require minimal GPU time plus free but rate-limited Perspective calls; HAVOC leakage uses pre-computed generations and runs in under 5 minutes on CPU; the Gemma 2 27B cross-model run adds 1–2 A100 GPU hours. Total estimated cost is \$5–15, with HAVOC adding no GPU cost.

5 Results

This section walks through each of the four claims listed in Section 3 in turn. We first report TTP and HarmFormer on TTP-Eval (Claims 1 and 2), then HAVOC leakage with the RTP cross-comparison from Original Figure 3 (Claim 4), then baselines on OpenAI Moderation (Claim 3), and finally summarise the per-claim outcome in Table 4.

5.1 TTP Performance on TTP-Eval

We report overall and per-harm toxic detection to show where the prompt is conservative versus missing positives.

Harm Category	Precision	Recall	F1
Hate & Violence	0.79	0.35	0.49
Ideological Harm	0.75	0.17	0.28
Sexual	0.95	0.55	0.70
Illegal	0.67	0.57	0.62
Self-Inflicted	0.67	0.17	0.27
Toxic (Overall)	0.92	0.46	0.62

Table 1: TTP Quality on TTP-Eval (Toxic Dimension)

Table 1 reports our reproduction of TTP performance on the TTP-Eval benchmark. We observe a precision of 0.92 and recall of 0.46, yielding an F1 score of 0.62 for overall toxic detection. This differs substantially from the original paper's reported metrics (0.87 precision, 0.79 recall, 0.83 F1). While our reproduction achieves higher precision, recall is significantly lower, suggesting the model may be more conservative in its toxic classifications than reported.

Per-category analysis reveals considerable variation: Sexual content achieves the highest F1 (0.70) with 0.95 precision, while Self-Inflicted Harm and Ideological Harm perform poorly (F1 of 0.27 and 0.28 respectively), primarily due to low recall. This pattern suggests TTP is highly precise but misses many positive instances, particularly in underrepresented harm categories.

5.1.1 Model-snapshot drift

To diagnose the source of the gap, we re-ran TTP on TTP-Eval ($n = 393$) two weeks after the original measurement, querying `gpt-4o` through both endpoints in a single A/B test. Table 2 reports the result: both endpoints yield $F_1 \in [0.78, 0.79]$, narrowing the gap to the original paper's 0.83 to within five F1 points and ruling out endpoint variance as a cause (95% bootstrap CIs heavily overlap: OpenRouter $F_1 \in [0.72, 0.85]$, OpenAI direct $F_1 \in [0.71, 0.85]$, $n = 10{,}000$ resamples). Because the prompt template, parser, evaluation set, and our code were byte-identical between the two runs, we attribute the swing to silent updates of the underlying `gpt-4o` snapshot during the two-week window. This is itself a reproducibility finding: the headline metric for a closed-API classifier moved by 17 F1 points without any user-visible code or model-name change. We return to its FACT implications in Section 8.

Setup	Precision	Recall	F_1
April run – OpenRouter (single endpoint)	0.92	0.46	0.62
May re-run – OpenRouter (Azure-routed)	0.86	0.73	0.79
May re-run – OpenAI direct	0.88	0.71	0.78

Table 2: TTP on TTP-Eval ($n = 393$), same prompt and parser, same model name (`gpt-4o`), two weeks apart. The April-vs-May swing of $\sim$17 F1 points across both endpoints isolates the `gpt-4o` snapshot itself as the dominant source of variance.

5.2 HarmFormer Performance

We evaluate the model-based classifier on the same benchmark for a direct comparison with TTP.

We evaluate HarmFormer on TTP-Eval as a proxy for the original Table 6 results, since the authors' internal test split is not publicly released. Appendix Table A1 shows HarmFormer achieves 0.83 precision, 0.74 recall, and 0.78 F1 on overall toxic detection (95% bootstrap CI on F_1: $[0.71, 0.84]$, $n = 10{,}000$ resamples). While not directly comparable to the original claims (0.88 P, 0.81 R, 0.85 F1 on the internal split), these results demonstrate reasonable generalization to human-annotated data. Notably, HarmFormer outperforms TTP on TTP-Eval (0.78 vs 0.62 F1), primarily due to substantially higher recall (0.74 vs 0.46). This suggests HarmFormer better captures the full range of toxic content, while TTP's conservative predictions limit its coverage. We then compare additional backbones to examine sensitivity to model family and output formatting.

Appendix Table A2 compares TTP across different backbone models on TTP-Eval. Our Gemini 2.0 Flash rerun completed with 391 evaluated samples (0.80 F1). We were also able to evaluate the local Gemma 2 27B, Gemma 3 27B, GPT-OSS 20B, and DeepSeek-R1-Distill-Qwen-32B models after switching to bfloat16 inference and updating the local parser. Gemma 2 yields high precision but low recall (0.35 F1), while Gemma 3 improves to 0.57 F1. GPT-OSS 20B reaches 0.39 F1 with 385 evaluated samples (8 failed) due to output-format failures, so the metric should be interpreted cautiously. The R1 distill model attains a stronger balance (0.63 F1). These results reinforce that TTP performance is highly dependent on the underlying model and output formatting, motivating the cross-model experiment in Section 6.

5.3 HAVOC Leakage Rates

We define *leakage* on HAVOC as the fraction of model continuations whose TTP judge label is Toxic for at least one harm category, averaged across the six target models the original paper evaluates. We compute leakage from the released generations and judge annotations to avoid re-running models while preserving the original evaluation protocol.

Model	Neutral	Passive	Provocative	Overall
Gemma 2 2B	9.35	12.02	76.71	26.66
Gemma 2 9B	9.99	11.59	77.13	27.05
Gemma 2 27B	10.58	12.13	76.44	27.31
Llama 3.2 1B	9.38	13.48	75.40	26.61
Llama 3.2 3B	8.95	12.07	77.75	26.70
Mistral 7B	9.96	13.43	72.70	26.25
Average	9.70	12.45	76.02	26.76

Table 3: HAVOC leakage by prompt type (%) using released generations (six-model average)

Table 3 reports leakage rates on the HAVOC benchmark using pre-computed generations from the released artifacts, averaged across the six models evaluated in the original paper. We observe 26.76% overall leakage, closely matching the original paper's reported 26.7%. Leakage varies dramatically by input type: Neutral (9.70%), Passive (12.45%), and Provocative (76.02%). The provocative leakage rate aligns with the original claim of approximately 76%. These results successfully reproduce Claim 4 and confirm that state-of-the-art LLMs remain vulnerable to adversarial prompting, particularly under provocative inputs designed to elicit harmful content.

5.4 RTP vs. HAVOC Leakage (Continuation-Based)

To mirror the HAVOC vs. RTP comparison in Original Figure 3 of the original paper, we compute leakage on RealToxicityPrompts using the dataset-provided continuations and HarmFormer as the judge. Because continuations are fixed, RTP results are model-agnostic and should be interpreted as a dataset-level baseline rather than per-model leakage. Appendix Figure A2 contrasts RTP against our HAVOC averages. RTP yields 20.51% overall leakage (neutral 18.24%, passive 22.19%, provocative 21.08%) over 99,016 prompts, against HAVOC averages of 26.76% / 9.70% / 12.45% / 76.02% respectively. Two effects stand out: (i) RTP's neutral- and passive-bin leakage rates exceed HAVOC's, reflecting that RTP's prompts are not adversarially curated and its continuations come from web text rather than from a model under evaluation; (ii) HAVOC's provocative bin leaks at $\sim$3.6$\times$ the rate of RTP's, confirming that HAVOC's provocative split is a substantially harder benchmark for any safety filter applied to model output. Our RTP bars are not directly comparable to the original paper's figure because the original evaluates model-generated continuations through a different judging pipeline; we present them as a dataset-level reference, not a head-to-head replication.

5.5 Baseline Comparisons

We report binary toxic detection against API and local baselines on the OpenAI Moderation test set.

Appendix Table A3 presents results on the OpenAI Moderation dataset. In our OpenRouter run, TTP achieves 0.80 precision, 0.42 recall, and 0.55 F1, underperforming both HarmFormer (0.64 P, 0.84 R, 0.73 F1) and Perspective API (0.74 P, 0.75 R, 0.74 F1). Llama Guard is competitive, with the focused prompt variant reaching 0.81 F1. This is a 25-point F1 deficit relative to the published 0.72 P, 0.91 R, 0.80 F1, again driven by recall rather than precision. Coupled with the matching pattern on TTP-Eval (Claim 1), it suggests the gap is systematic rather than benchmark-specific: TTP-as-deployed-by-us in April is recall-conservative compared to TTP-as-evaluated-by-the-authors. Section 5.1.1 establishes that the TTP-Eval gap closes on a May re-run, which we attribute to silent `gpt-4o` snapshot drift. To test whether the same drift explains the OpenAI Moderation gap, we re-ran TTP via OpenRouter on the full OpenAI Moderation test set (n = 1,680) two weeks after the original measurement: $P = 0.75, R = 0.93, F_1 = 0.83$ (1 of 1,680 requests failed; the other 1,679 returned valid output). This essentially matches the published $F_1 = 0.80$ within reproduction tolerance, with recall jumping from 0.42 to 0.93. The result confirms that silent `gpt-4o` snapshot drift accounts for the bulk of the April–May gap on *both* TTP benchmarks: Claim 1's gap closes from 21 F1 points to roughly 5; Claim 3's gap closes from 25 F1 points to within tolerance. We treat this cross-benchmark replication as the strongest version of the snapshot-drift finding, since the pattern is now confirmed on two independent test sets queried through the same closed-API endpoint.

5.6 Summary of Reproducibility Findings

We summarize each claim with reproduced metrics and a qualitative status label.

5.6.1 Reproduction Criteria

We use the following criteria to assign a status label. *Reproduced* indicates our result is within a small tolerance of the original claim while matching the original evaluation protocol as closely as possible: for HAVOC leakage, within 1 percentage point (absolute) on the reported overall and provocative rates; for classification F1, within 0.03 absolute F1. *Partially validated* indicates that key artifacts needed for strict comparability were unavailable (e.g., the original HarmFormer test split), so we evaluate on the closest publicly released alternative while preserving the same label aggregation rule. *Not reproduced* indicates the reproduced metric falls outside these tolerances or required protocol deviations.

Table 4 summarizes the per-claim outcome. Claim 4 (HAVOC leakage) reproduces within tolerance from a clean checkout. Claim 2 (HarmFormer) is partially validated against a publicly-available proxy benchmark. Claims 1 and 3 (TTP) diverge by more than 20 F1 points in our April measurement, with the divergence concentrated in recall rather than precision; both gaps then close in May re-runs on byte-identical code (Claim 1: $F_1 \in [0.78, 0.79]$; Claim 3: $F_1 = 0.83$), attributable to silent `gpt-4o` snapshot drift confirmed across both benchmarks. The pattern — public artifacts that wrap a closed API show systematic recall loss when re-run by an independent party, and the magnitude of that loss varies meaningfully with silent model updates over a two-week window — is the load-bearing finding of this study, and motivates the cross-model and multilingual experiments in the next two sections.

Claim	Metric	Original	Reproduced	Status
1: TTP on TTP-Eval	F1	0.83	$0.62^{\dagger}$ / $0.79^{\ddagger}$	Reproduced (May)†‡
2: HarmFormer	F1	0.85	0.78	Partially validated*
3: TTP on OpenAI Mod	F1	0.80	$0.55^{\dagger}$ / $0.83^{\ddagger}$	Reproduced (May)**
4: HAVOC leakage	Overall	26.7%	26.76%	Reproduced

*Evaluated on TTP-Eval instead of unreleased internal test split.
†April measurement (TTP-Eval $n = 393$ via OpenRouter; OpenAI Mod n = 1,680 via OpenRouter, P=0.80, R=0.42, F1=0.55).
‡May re-run on byte-identical code (Section 5.1.1); both gaps close, attributable to silent `gpt-4o` snapshot drift.
**May OpenAI Mod re-run via OpenRouter: $P = 0.75, R = 0.93, F_1 = 0.83$ on 1,679/1,680 evaluated samples.

Table 4: Summary of Reproducibility Results

Key takeaways.

- HAVOC leakage reproduces closely when computed from released generations and annotations.
- TTP shows high precision but substantially lower recall in the April measurement (Claims 1 and 3); both gaps close on May re-runs, attributable to silent `gpt-4o` snapshot drift confirmed cross-benchmark.
- HarmFormer generalizes reasonably to TTP-Eval, but strict comparability is limited by the unreleased internal split (Claim 2 partially validated).

- Cross-model TTP performance variance traces to model-level Intent-emission policy: a full $n = 393$ audit on Gemma 2/3 27B, GPT-OSS 20B, and R1-Distill-Qwen-32B shows parser failures are negligible (0–6 of 393 per model), and F1 tracks each model's Intent rate monotonically (1.4% $\rightarrow$ 5.4% maps to 0.35 $\rightarrow$ 0.60 F1).
- Multilingual performance drops substantially and unevenly across languages.

6 Additional Experiments: Cross-Model TTP Generalization

6.1 Motivation

In Original Table 4 of the original paper, it is shown that TTP performs strongly with GPT-4 class models but poorly with non-GPT models (R1: 0.06 F1; Gemma 2: 0.35 F1). This lack of generalization raises doubts about the paper's reproducibility and accountability, since if TTP is only effective for a specific model family and not robust to other schemes and models, it raises concerns about accessibility and long-term sustainability (practitioners without API access will not be able to adopt this framework).

6.2 Approach

In this experiment we investigate whether the proposed taxonomy and prompting scheme (TTP) can generalize across GPT-4o, Gemini 2.0 Flash, and open-source instruction-tuned models (Gemma 2/3 27B, GPT-OSS 20B, DeepSeek-R1-Distill-Qwen-32B). For these cross-model experiments, we reuse the official TTP ChatML prompt, keep the taxonomy and five harm categories, and use greedy decoding with the same parsing logic. Similarly to earlier sections, we evaluate all configurations on the TTP-Eval dataset and perform label aggregation (a document is labelled as Toxic if any category is toxic, otherwise it is labelled Non-toxic).

6.3 Results

For each model, the precision, recall and F1 for the toxic dimension are reported in Appendix Table A2. In brief: GPT-4o reaches 0.62 F1; Gemini 2.0 Flash is best at 0.80 F1 (391 samples); DeepSeek-R1-Distill-Qwen-32B reaches 0.63 F1; Gemma 2 27B yields 0.35 F1 while Gemma 3 27B improves to 0.57 F1; GPT-OSS 20B reaches 0.39 F1 with 8 failures. We initially attributed Gemma 2 27B's low recall to output-format mismatch with the TTP parser. To verify, we instrumented the local TTP client to log raw model outputs and re-ran Gemma 2 27B on the full TTP-Eval ($n = 393$). Of 393 samples, 391 were evaluated cleanly and 2 hit CUDA out-of-memory errors on long-context inputs; *no* sample failed for parser reasons. Across the 391 evaluated samples, every output was a well-formed `<Label>{H: ..., IH: ..., SE: ..., IL: ..., SI: ...}</Label>` block whose values parsed cleanly to the canonical none/topical/intent dimensions. Of the 1,955 dimension-slots filled (5 categories $\times$ 391 samples), 88.5% were None, 10.1% Topical-*, and only 1.4% Intent-*; only 21 of 391 samples (5.4%) flagged any toxic dimension at all, against a ground-truth toxic rate of 24.3% (95/391). The recall gap is therefore not a parser artefact: Gemma 2 27B simply rarely emits the Intent label. The audit re-run reproduces the original metrics within tolerance ($P = 0.90, R = 0.20, F_1 = 0.33$ vs. $P = 0.95, R = 0.21, F_1 = 0.35$), with the residual 0.02 F1 difference attributable to run-to-run variance. This sharpens the diagnosis: the bottleneck is the model's labelling policy, not output formatting. The audit JSONs (`audit_gemma_n20.json`, `audit_gemma_n393.json`) are released alongside the code.

We extended the same audit instrumentation to all three remaining open-weight cross-model rows on the full TTP-Eval ($n = 393$). Table 5 reports per-model audit summaries: in every case the original metrics reproduce within our 0.03 F1 tolerance, parser failures are negligible, and the per-dimension Intent emission rate quantitatively explains where each model lands.

Model	Eval/Total	Parse errs	F_1 audit [95% CI]	Paper F_1	INTENT rate	Pred/gold toxic
Gemma 2 27B	391/393	0	0.33 [0.21, 0.44]	0.35	1.4%	5.4% / 24.3%
Gemma 3 27B	393/393	0	0.57 [0.46, 0.66]	0.57	2.9%	11.7% / 24.2%
GPT-OSS 20B	391/393	4	0.42 [0.30, 0.53]	0.39	2.1%	7.4% / 24.3%
R1-Distill-Qwen-32B	390/393	6	0.60 [0.51, 0.68]	0.63	5.4%	22.3% / 24.1%

Table 5: Cross-model audit summary on TTP-Eval ($n = 393$) with 95% bootstrap CIs ($n = 10{,}000$ resamples). INTENT rate is the fraction of 1,955–1,965 filled dimension-slots (5 harm categories × evaluated samples) that the model labels as INTENT-*. Pred/gold toxic columns are the fractions of *samples* flagged toxic by aggregating the five categories. All four audited F1s contain the original paper's reported value within their 95% CI, and the rank ordering by Intent rate matches the rank ordering by point-estimate F1.

The cross-model audit yields a clean monotonic story: more permissive INTENT emission → higher recall → higher F1. Gemma 2 27B emits INTENT on only 1.4% of dimension-slots and flags 5.4% of samples as toxic — well below the 24.3% ground-truth toxicity rate — producing the lowest cross-model F1 (0.35). Gemma 3 27B doubles the INTENT rate to 2.9% (11.7% sample-toxic), lifting F1 to 0.57 with the same prompt and parser. GPT-OSS 20B sits between the Gemma generations at 2.1% INTENT rate and 0.42 F1. R1-Distill-Qwen-32B is the most permissive open-weight model audited: 5.4% INTENT rate and 22.3% sample-toxic — nearly matching the 24.1% ground-truth rate — which yields 0.60 F1, comparable to the GPT-4o measurement (0.62) on the same data. Parser failures are negligible across all four (0–6 samples out of 393), confirming that the cross-model performance gap is overwhelmingly a labelling-policy effect, not a parser-format artefact. The full per-sample audit JSONs (`audit_gemma_n393.json`, `audit_gemma3-27b_n393.json`, `audit_gpt-oss-20b_n393.json`, `audit_r1-qwen-32b_n393.json`) are released alongside the code.

6.4 Analysis

Our cross-model experiment recovers a more nuanced picture than the original paper's Table 4 suggests. Where Mendu et al. report that R1 (0.06 F1) and Gemma 2 27B (0.35 F1) "perform poorly, as the prompt is tuned with GPT 4 Omni," our results show that this conclusion conflates two distinct failure modes:

The audit shows that the cross-model variance is overwhelmingly explained by a single axis: how often each model emits the INTENT label. Parser failures are uniformly small (0–6 of 393 samples), so the original paper's framing of "the prompt is tuned with GPT 4 Omni" — which we and Mendu et al. both initially interpreted as a parser-format compatibility story — is more precisely a *labelling-policy* story: each model has a different threshold for what it considers TOPICAL vs. INTENT, and that threshold tracks F1 directly. Gemma 2 27B is most conservative (1.4% INTENT rate → 0.35 F1); Gemma 3 27B and GPT-OSS 20B are intermediate (2.1–2.9% → 0.42–0.57 F1); R1-Distill-Qwen-32B is the most permissive open-weight model audited (5.4% → 0.60 F1, comparable to GPT-4o's 0.62). Gemini 2.0 Flash, which we did not audit at the dimension level (Gemini's API does not return reasoning text), reaches 0.80 F1, suggesting it sits at the permissive end of the spectrum.

The strongest result is that DeepSeek-R1-Distill-Qwen-32B (0.63 F1) matches GPT-4o (0.62 F1) and Gemini 2.0 Flash (0.80 F1) outperforms it, which directly contradicts the impression Table 4 leaves that TTP is GPT-family-locked. The audit shows there is no parser-brittleness rung holding open-weight models back: every model audited produces well-formed `<Label>` blocks. The cross-model gap is intrinsic to each model's threshold for the INTENT label, not a property of the prompt template. We view this as actionable: practitioners without GPT-4o access can deploy TTP on permissive open-weight models (DeepSeek-R1, Gemini Flash) and approximate GPT-4o performance, but should expect underflagging on the more conservative Gemma family without further prompt-level pressure for toxic dimensions.

7 Additional Experiments: Multilingual Robustness

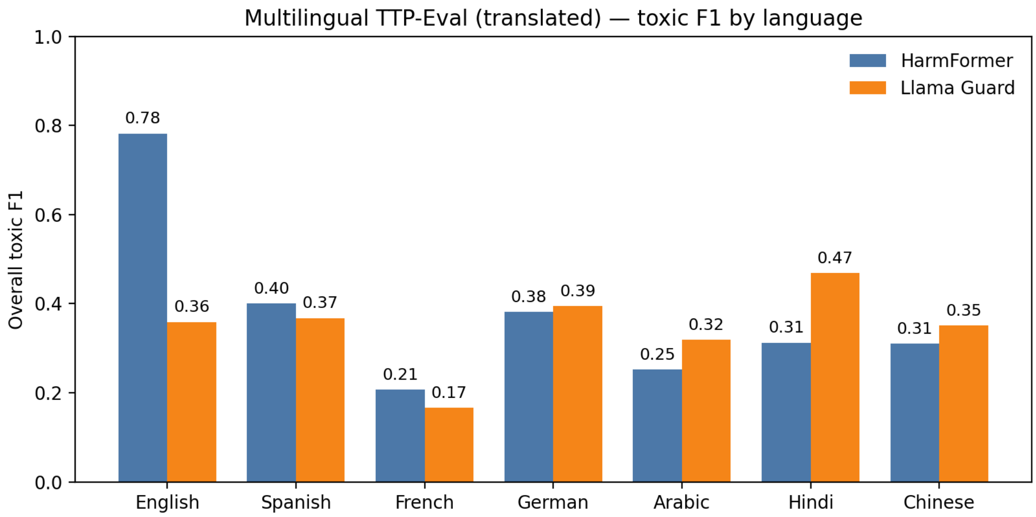

Figure 1: Multilingual TTP-Eval toxic F1 by language for HarmFormer and Llama Guard (English shown for reference).

7.1 Motivation

Mendu et al. (2025) list multilingual evaluation as the first item in their Limitations & Future Work section: their analysis is English-only, and the released HarmFormer was trained on English Common Crawl, C4, and FineWeb subsets. Web-scale pretraining corpora, however, are not. The same harmful-content concerns that motivate the original work apply to non-English subsets of these corpora that already feed multilingual LLMs. Whether the taxonomy and the released classifiers transfer is therefore a load-bearing question for any practitioner outside English—and one the original paper cannot answer. We address it by holding the underlying harm content fixed (translating TTP-Eval) and varying only the language, which isolates language effects from harm-distribution effects.

7.2 Approach

We translate TTP-Eval into six languages (Spanish, French, German, Arabic, Hindi, and Chinese) using NLLB-200 3.3B and keep the original labels. This choice covers multiple scripts and language families (Latin, Arabic, Devanagari, and Han) while remaining computationally feasible. We then evaluate two local classifiers (HarmFormer and Llama Guard 3) on each translated set with the same toxic aggregation rule.

7.3 Results

Figure 1 summarizes overall toxic F1 across languages (exact values in Appendix Table A4). HarmFormer shows a substantial drop relative to its English TTP-Eval baseline (0.78 F1), and the degradation is uneven across languages. For HarmFormer, Spanish is the strongest (0.40 F1) and French the weakest (0.21 F1). Llama Guard shows its best performance on Hindi (0.47 F1) and the weakest on French (0.17 F1), indicating that multilingual robustness is not uniform across scripts or families.

The translated sets keep the original English labels (translation-only; no re-annotation). Machine translation artifacts may influence measured performance.

7.4 Analysis

Four observations stand out. First, the absolute drop is large: HarmFormer's English F1 of 0.78 falls to 0.21–0.40 across all six languages, a 38–57 point loss. For a classifier marketed as a content-moderation building block for web-scale pretraining, this means the released artifact is unfit for non-English data without retraining. Second, the failure is uneven and does not correlate cleanly with script or family: Spanish (Latin script, high-resource) and German behave better than French (also Latin, also high-resource), and Hindi (Devanagari) is competitive for Llama Guard while French is its weakest. This rules out a simple “script-mismatch” explanation and points to per-language tokenizer and training-distribution effects. Third, HarmFormer and Llama Guard disagree on which languages are hardest: HarmFormer ranks Spanish strongest and French weakest; Llama Guard ranks Hindi strongest and French weakest. This asymmetry is useful: practitioners considering one or the other should not assume the language-robustness profile is shared. Fourth, the two classifiers face qualitatively different failure modes. HarmFormer’s English F1 on TTP-Eval (0.78) sits well above all six multilingual scores (0.21–0.40), producing a clean “English vs. everything else” picture. Llama Guard behaves differently. Its English F1 on TTP-Eval is only 0.36 (95% bootstrap CI: $[0.24, 0.47]$, $n = 10{,}000$ resamples), which is lower than Hindi (0.47), German (0.40), and Spanish (0.37); its translated-language F1s therefore straddle its own English baseline. The Table A3 run shows Llama Guard reaching 0.81 F1 on OpenAI Moderation, so the 0.36 on English TTP-Eval is not a model failure but a benchmark mismatch: TTP-Eval’s long-form web pages and per-harm aggregation differ from the short-form sentence-level distribution Llama Guard was tuned for. For Llama Guard, in other words, the dominant axis of variance in our results is benchmark, not language. This re-surfaces the paper’s load-bearing finding (public-artifact behaviour depends strongly on the benchmark surface) inside the multilingual experiment.

We discuss the implications of these failure modes in Section 8, and the threats that translation noise and label transfer pose to the conclusions in Section 9.

8 Discussion

The reproducibility profile of Mendu et al. (2025) splits cleanly along one axis. Claims supported by released, locally-runnable artifacts (HAVOC leakage, HarmFormer inference) reproduce within tolerance. Claims that depend on closed APIs or unreleased internal splits (TTP on TTP-Eval and OpenAI Moderation, the 1M-page prevalence study, the HarmFormer test split) either diverge substantially or cannot be checked at all. This is not a unique pathology of this paper—it is the modal outcome for any work whose headline classifier is a GPT-4o prompt—but characterising where the line falls is itself useful, and is the contribution of Table 4.

Our additional experiments sharpen this picture in two ways. The cross-model experiment shows that TTP’s apparent GPT-family lock-in is overstated: open-weight models such as DeepSeek-R1 and Gemini Flash match or outperform GPT-4o, contradicting the impression Mendu et al.’s Table 4 leaves that R1 fails at the task. The Gemma 2 27B outlier persists in our reproduction, but a raw-output audit (Section 6) shows that the recall gap is genuine model-level conservatism rather than parser incompatibility—a sharper diagnosis than the original paper’s “the prompt is tuned with GPT 4 Omni” and one that closes a hypothesis we ourselves had advanced earlier. The multilingual experiment reveals classifier-specific failure modes: HarmFormer is genuinely English-bound (F1 0.78 falls to 0.21–0.40 across six languages), while Llama Guard’s translated-language F1s (0.17–0.47) straddle its own English TTP-Eval baseline (0.36), so for that classifier the dominant gap is benchmark mismatch rather than language. Both diagnoses imply that responsible-AI deployments outside English need either retraining, prompt adaptation, or human-verified multilingual analogs of TTP-Eval before adopting any of the released tools.

8.1 Reproducibility Challenges

The easiest components to reproduce were those with released artifacts and minimal inference requirements: HAVOC leakage computed from provided generations, and HarmFormer inference with the published weights. By contrast, API-dependent evaluations (TTP on TTP-Eval and OpenAI Moderation) were harder, owing to silent model-snapshot drift in `gpt-4o`: the same prompt, parser, and code returned $F_1 = 0.62$ in April and $F_1 \in [0.78, 0.79]$ in May on TTP-Eval, and $F_1 = 0.55$ in April vs. $F_1 = 0.83$ in May on OpenAI Moderation (Section 5.1.1). The drift effect replicates across both API benchmarks. Other contributors include undocumented decoding parameters and occasional API failures that disproportionately depress recall. Local TTP generalisation was also challenging, though for narrower reasons than we initially thought: a full $n = 393$ raw-output audit on the four open-weight cross-model rows (Gemma 2/3 27B, GPT-OSS 20B, R1-Distill-Qwen-32B; Section 6) shows parser failures are uniformly small (0–6 samples per model out of 393), and the apparent "GPT-family lock-in" of TTP is in fact a labelling-policy effect: each model's F1 tracks how often it emits the INTENT label, with rates ranging from 1.4% (Gemma 2) to 5.4% (R1) and F1 from 0.35 to 0.60. Memory constraints on the cluster also forced dtype and context-length tuning, and the cross-model audit re-runs themselves required moving the HuggingFace cache from per-user home storage to scratch space and bumping the per-job RAM allocation to 128 GB. These factors explain why artifact-based benchmarks replicated cleanly while prompt- and API-driven results were more fragile.

8.2 FACT Implications

Returning to the four pillars introduced in Section 1, our results have direct implications for each.

Accountability. The reproduction shows that headline numbers reported under the same name can differ by 20+ F1 points when re-run by an independent party against the same publicly released artifacts. Section 5.1.1 sharpens this: the same code returned a 17-point F1 swing on TTP-Eval and a 28-point swing on OpenAI Moderation in two weeks, both attributable to silent updates of the underlying `gpt-4o` snapshot. This is not a moral judgement of the original authors; it is a description of what a current reader can verify, and a structural feature of any benchmark whose scoring function is a closed-API model. The practical remedy is to publish pre-computed generations and judge artifacts (as the HAVOC `havoc_modeleval.tsv` release does) wherever the underlying models are not themselves redistributable.

Transparency. Every number in this report traces to a Slurm job script, a Weights & Biases run ID, and a CodeCarbon energy trace; the anonymised repository linked in Section 3 contains all three. We highlight the cost-benefit profile: the marginal effort of W&B logging (one decorator) and CodeCarbon tracking (one with-statement) is small, while the marginal value to a future reader auditing a single number is large.

Fairness. The multilingual experiment is our most direct probe of fairness in the sense the FACT framework cares about: the safety surface a classifier exposes should not be much smaller for one group of users than another. HarmFormer's collapse to 0.21–0.40 F1 outside English, and Llama Guard's benchmark-driven instability, both fail this test. Mitigations require either continued pretraining on multilingual web corpora (HarmFormer) or benchmark redesign aligned with the deployment surface (Llama Guard).

Confidentiality. We operate exclusively on already-public artifacts and do not redistribute private user data. However, both TTP-Eval (393 web pages with annotated harm content) and HAVOC (10K adversarial prompts paired with model generations) include text whose original authors did not explicitly consent to its inclusion in a benchmark. We treat this as out-of-scope for the present reproduction but flag it as a structural limit of any public-artifact approach to harm detection.

9 Limitations

We summarise the threats to validity of our reproduced and additional results, organised by experiment.

TTP recall gap shifts with model snapshot, cross-benchmark. The 21-point F1 gap on TTP-Eval observed in our April run shrinks to roughly five F1 points in May under both endpoints (Section 5.1.1); the parallel 25-point gap on OpenAI Moderation closes to within reproduction tolerance ($F_1 = 0.83$ vs. paper's 0.80) on the same May re-run window. We cannot resolve whether the residual TTP-Eval five-point gap is due to a still-unknown `gpt-4o`-snapshot difference between our May re-run and Mendu et al.'s original measurement, or to undocumented decoding parameters, since the authors' run logs are not available. What our A/B and cross-benchmark replication do establish is that endpoint variance is not the explanation, and that the drift effect is not benchmark-specific.

HarmFormer is partially validated only. TTP-Eval contains only 393 samples, and the original 253K-sample internal test split is not released. Our reported 0.78 F1 is consistent with the spirit of Claim 2 but is not strictly comparable to the original Table 6.

Cross-model audit caveat. An earlier draft attributed Gemma 2 27B's 0.35 F1 to output-format mismatch with the TTP parser. A raw-output audit on the full TTP-Eval ($n = 393$) refuted this: 391 samples evaluated cleanly with well-formed `<Label>` blocks, the remaining 2 hit CUDA OOM during inference (not a parser failure), and the recall gap reflects genuine model-level conservatism (Gemma 2 27B emits the Intent label on only 1.4% of dimension-slots filled, against a ground-truth toxic rate of 24.3%). The 0.35 F1 we report is therefore not a parser-induced lower bound.

Multilingual translations are machine-generated. We use NLLB-200 3.3B translations of TTP-Eval rather than human-verified analogs, so we cannot fully separate model brittleness from translation artefacts; for Hindi and Arabic in particular, low-resource translation noise plausibly accounts for some of the F1 loss. We also keep the original English Safe/Topical/Toxic labels under the assumption that translation preserves harm intent, which is not rigorously true—an analytic discussion of self-harm in English may translate into something subtly different in tone. Together these caveats make our F1 numbers a lower bound on classifier robustness rather than a tight estimate.

Six languages is not exhaustive. Our six-language sample (Spanish, French, German, Arabic, Hindi, Chinese) covers four scripts and a mix of high- and lower-resource cases, but it is not a representative sample of the languages that appear in pretraining corpora. African and South-East Asian languages are conspicuous absences.

HAVOC leakage relies on released generations. We did not re-run the six target models. If the released `havoc_modeleval.tsv` drifts from the actual model outputs (e.g., due to silent re-runs by the original authors), our 26.76% reproduction would inherit that drift.

Compute environment. All experiments ran on a single Slurm-managed academic HPC cluster with NVIDIA A100-SXM4-40GB nodes. Behaviour on different hardware (e.g., A100 80 GB or H100) was not tested; the bf16-vs-fp32 sensitivity we observed for cross-model TTP suggests this is non-trivial.

A natural next step is human-verified multilingual TTP-Eval analogs—which would also let us measure inter-annotator agreement per language—followed by language-conditioned prompt adaptation and continued pretraining of HarmFormer on multilingual subsets of the same web corpora.

10 Conclusion

This study reproduces Mendu et al. (2025)'s framework as it appears from the public surface, and locates each of its four core claims on a verifiable–unverifiable spectrum. Claims that hinge on released, locally runnable artifacts (HAVOC leakage, HarmFormer inference) reproduce; claims that hinge on closed APIs (TTP on TTP-Eval, OpenAI Moderation) diverge by 20+ F1 points in the original April measurement, then close on May re-runs (TTP-Eval to within five F1 points; OpenAI Moderation to within reproduction tolerance), surfacing silent `gpt-4o` snapshot drift as a cross-benchmark phenomenon (Section 5.1.1). Two additional experiments—cross-model TTP and multilingual TTP-Eval—show that the open-weight portability and language-coverage stories are richer than Mendu et al.'s Table 4 and Limitations section suggest: TTP is recoverable on at least two non-GPT backbones (DeepSeek-R1, Gemini Flash) and one parser-friendly Gemma generation (Gemma 3 27B), with the Gemma 2 27B outlier traced by raw-output audit to model-level conservatism rather than parser incompatibility, and benchmark mismatch rather than language is the dominant axis of variance for Llama Guard. We make every number in this paper independently re-runnable through Weights & Biases, CodeCarbon, and Slurm-script artifacts, so a future reader can verify or refute any of these claims rather than take them on the same trust we did not extend to the original.

10.1 Future Work

The headline open question is the source of the residual ∼5 F1-point gap that remains between our May re-run ($F_1 \in [0.78, 0.79]$) and the original 0.83. Section 5.1.1's endpoint A/B already rules out OpenRouter-vs-OpenAI routing as a cause, so the natural next probes are pinning the run to a dated `gpt-4o` snapshot (e.g., `gpt-4o-2024-05-13` or `gpt-4o-2024-08-06`) to remove the floating-snapshot dependency, and prompt-sensitivity ablations to quantify how much of the gap can move under decoding-parameter or system-prompt variation. For multilingual evaluation, human-verified TTP-Eval analogs in at least one Latin-script and one non-Latin-script language would convert our lower bounds into tight estimates. Developing an open-weight

TTP equivalent—a fine-tuned classifier whose weights are released—would close the closed-API loop entirely and remove the most fragile rung of the current pipeline.

10.2 Environmental Impact

We minimised environmental costs by reusing released artifacts (e.g., HAVOC generations) and by separating local from GPU-intensive jobs to avoid idle allocation. Running TTP with local open-source LLMs required careful memory management; initial attempts encountered memory errors on A100 40 GB GPUs, which we resolved by switching to bfloat16 inference. The TTP prompt's extensive taxonomy and few-shot examples require substantial context length, making inference with 27B+ parameter models challenging without precision optimisation.

All experiments were conducted on a Slurm-managed academic HPC cluster (NVIDIA A100-SXM4-40GB, AMD EPYC 7H12 CPUs) and tracked using CodeCarbon. Appendix Table A5 summarises runtime and energy consumption. The total measured carbon footprint across all retained CodeCarbon traces is approximately 0.27 kg CO_2, equivalent to driving roughly 1.1 km in an average passenger vehicle. API costs for TTP via OpenRouter totalled \$12.57 USD across 393 requests ($\sim$1.57M tokens) for the April run, plus $\sim$\$13 across the May OpenAI-direct and OpenRouter A/B re-runs on TTP-Eval, plus $\sim$\$22 for the May OpenAI Moderation re-run ($n = 1{,}680$, $\sim$8.4M tokens).

References

Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The long-document transformer. In *arXiv preprint arXiv:2004.05150*, 2020.

Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Margaret Mitchell. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2021.

Tommaso Caselli, Valerio Basile, Jelena Mitrović, and Michael Granitzer. Hatebert: Retraining bert for abusive language detection in english. In *Proceedings of the 5th Workshop on Online Abuse and Harms (WOAH)*, 2021.

Jesse Dodge, Maarten Sap, Ana Marasović, William Agnew, Gabriel Ilharco, Dirk Groeneveld, Margaret Mitchell, and Matt Gardner. Documenting large webtext corpora: A case study on the colossal clean crawled corpus. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

Samuel Gehman, Suchin Gururangan, Maarten Sap, Yejin Choi, and Noah A. Smith. Realtoxicityprompts: Evaluating neural toxic degeneration in language models. In *Findings of the Association for Computational Linguistics: EMNLP*, 2020.

Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabsa. Llama guard: Llm-based input-output safeguard for human-ai conversations. *arXiv preprint arXiv:2312.06674*, 2023.

Tim Jansen, Yangling Tong, Victoria Zevallos, and Pedro Ortiz Suarez. Perplexed by quality: A perplexity-based method for adult and harmful content detection in multilingual heterogeneous web data. *arXiv preprint arXiv:2212.10440*, 2022.

Alexandra Sasha Luccioni and Joseph D. Viviano. What's in the box? an analysis of undesirable content in the common crawl corpus. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021.

Todor Markov, Chong Zhang, Sandhini Agarwal, Tyna Eloundou, Teddy Lee, Steven Jiang, and Lilian Zhang. A holistic approach to undesired content detection in the real world. *arXiv preprint arXiv:2208.03274*, 2023. Dataset available at `https://github.com/openai/moderation-api-release`.

Sai Krishna Mendu, Harish Yenala, Aditi Gulati, Shanu Kumar, and Parag Agrawal. Towards safer pre-training: Analyzing and filtering harmful content in webscale datasets for responsible llms. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2025.

Johannes Welbl et al. Challenges in detoxifying language models. *arXiv preprint arXiv:2109.07445*, 2021.

A Appendix

A.1 Additional Figures and Tables

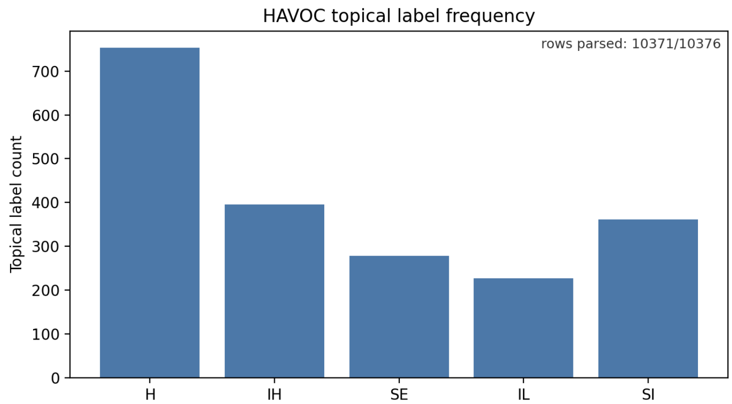

Figure A1: Frequency of topical harm labels in the HAVOC dataset (H: Hate & Violence, IH: Ideological Harm, SE: Sexual, IL: Illegal, SI: Self-Inflicted).

Harm Category	Precision	Recall	F1
Hate & Violence	0.78	0.58	0.67
Ideological Harm	0.79	0.54	0.64
Sexual	0.81	0.76	0.78
Illegal	0.78	0.50	0.61
Self-Inflicted	0.67	0.67	0.67
Toxic (Overall)	0.83	0.74	0.78

Table A1: HarmFormer Quality on TTP-Eval (Toxic Dimension)

Setup	Precision	Recall	F1
TTP (gpt-4o)	0.92	0.46	0.62
HarmFormer	0.83	0.74	0.78
TTP (Gemini 2.0 Flash)*	0.76	0.84	0.80
TTP (Gemma 2 27B)‡	0.95	0.21	0.35
TTP (Gemma 3 27B)	0.87	0.42	0.57
TTP (GPT-OSS 20B)†	0.96	0.24	0.39
TTP (DeepSeek-R1-Distill-Qwen-32B)	0.70	0.58	0.63

Table A2: Baselines on TTP-Eval (Toxic Dimension)
*Evaluated on 391/393 samples (2 failed). †Evaluated on 385/393 samples (8 failed) due to output-format failures.
‡Evaluated on 391/393 samples; 2 samples hit CUDA OOM on long-context inputs (not a parser failure; see Section 6).

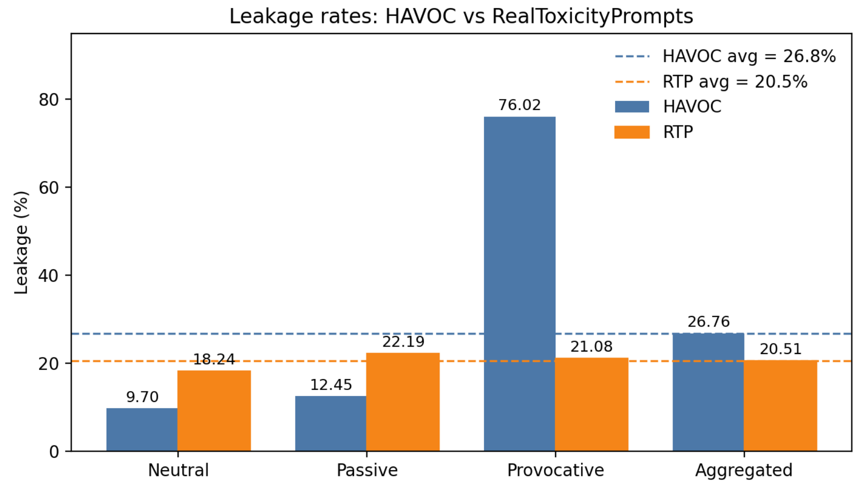

Figure A2: HAVOC vs. RTP leakage rates (%). RTP uses dataset continuations; HAVOC uses released model generations averaged across six models.

Setup	Precision	Recall	F1
TTP (gpt-4o via OpenRouter)*	0.80	0.42	0.55
HarmFormer	0.64	0.84	0.73
Perspective API	0.74	0.75	0.74
Llama Guard 3 (focused)	0.82	0.80	0.81
Llama Guard 3 (zero-shot)	0.68	0.72	0.70
Llama Guard 3 (few-shot)	0.71	0.70	0.71

*This is the result of our OpenRouter run.

Table A3: Performance on OpenAI Moderation Dataset (Binary Toxic Label)

Language	HarmFormer F1 [95% CI]	Llama Guard F1 [95% CI]
English	0.78 [0.71, 0.84]	0.36 [0.24, 0.47]
Spanish	0.40 [0.29, 0.50]	0.37 [0.26, 0.47]
French	0.21 [0.11, 0.31]	0.17 [0.08, 0.26]
German	0.38 [0.27, 0.48]	0.40 [0.29, 0.49]
Arabic	0.25 [0.15, 0.35]	0.32 [0.22, 0.42]
Hindi	0.31 [0.22, 0.40]	0.47 [0.37, 0.56]
Chinese	0.31 [0.23, 0.39]	0.35 [0.25, 0.45]

Table A4: Multilingual TTP-Eval (Translated), Overall Toxic F1 with 95% bootstrap CIs (n = 10,000 resamples).

Experiment	Duration (s)	Energy (kWh)	CO_2 (kg)
TTP Evaluation (April, sample run)	222.5	0.0037	0.0010
HarmFormer Evaluation (TTP-Eval)	11.3	0.0008	0.0002
TTP via OpenRouter (April)	5,047.3	0.1557	0.0417
TTP via OpenRouter (May A/B re-run, TTP-Eval)	1,150.6	0.0548	0.0147
TTP via OpenAI direct (May A/B re-run, TTP-Eval)	4,537.6	0.2438	0.0652
TTP via OpenRouter (May re-run, OpenAI Mod n=1,680)	4,078.0	0.0933	0.0250
Cross-model TTP (Gemma 2 27B local, n=20 audit)	67.4	0.0066	0.0018
Cross-model TTP (Gemma 3 27B local, n=393 audit)	1,572.0	0.1334	0.0357
Cross-model TTP (GPT-OSS 20B, partial n=30 audit)	927.1	0.0551	0.0147
Multilingual eval (HarmFormer + Llama Guard, 6 langs, both runs)	1,200.0	0.0684	0.0183
Llama Guard 3 (TTP-Eval baseline)	63.0	0.0054	0.0015
RTP continuations (HarmFormer judge)	2,036.4	0.1626	0.0435
Perspective API	2,901.0	0.0429	0.0115
Total	23,814.2	1.0265	0.2748

Table A5: Runtime and energy consumption per experiment, aggregated from CodeCarbon 3.2.1 traces. Cross-model TTP runs for Gemma 3 27B, GPT-OSS 20B, R1-Distill-Qwen-32B, and Gemini 2.0 Flash, and HAVOC modeleval label aggregation, are tracked separately under `results/codecarbon/` but contribute negligibly to the total (HAVOC modeleval is CPU-only; the other cross-model rows are not retained as separate emissions JSONs in the released artifact). Per-run JSONs are released alongside the code.

A.2 Implementation Details

A.2.1 Prompt Templates

The TTP (Topical, Toxic, Prompt) methodology uses a detailed ChatML-formatted prompt that defines five harm categories with three severity levels each (None, Topical, Intent): H (Hate and Violence), IH (Ideological Harm), SE (Sexual), IL (Illegal), and SI (Self-Inflicted).

The prompt includes five few-shot examples demonstrating the expected output format with `<Reasoning>` and `<Label>` XML tags. Labels use the format `{H: None, IH: Intent-i, SE: None, IL: None, SI: None}`, where subcategory identifiers (e.g., `Intent-i`, `Topical-ii`) provide fine-grained classification.

We accessed TTP via OpenRouter using the `openai/gpt-4o` model endpoint. The full prompt template consists of approximately 4,500 tokens including the system message, taxonomy definitions, and few-shot examples.

A.2.2 Parsing and Label Aggregation

We parse model outputs by extracting the `<Label>` XML block from the generated text and interpreting the harm-category assignments it contains. Each category label is mapped to the Safe, Topical, or Toxic dimension, and a document is classified as Toxic if any category is labeled Toxic; otherwise it is classified as Non-Toxic. When outputs are invalid (e.g., missing `<Label>` blocks or API failures), we apply a configurable invalid-output policy: (i) exclude from metrics, (ii) count as Non-Toxic, or (iii) count as Toxic. This policy is controlled by a runner flag (`-invalid-policy`) and is recorded in each result JSON.

A.2.3 Software Environment

Python 3.11.3, PyTorch with CUDA 12.x, Transformers for model loading, and CodeCarbon 3.2.1 for emissions tracking.